



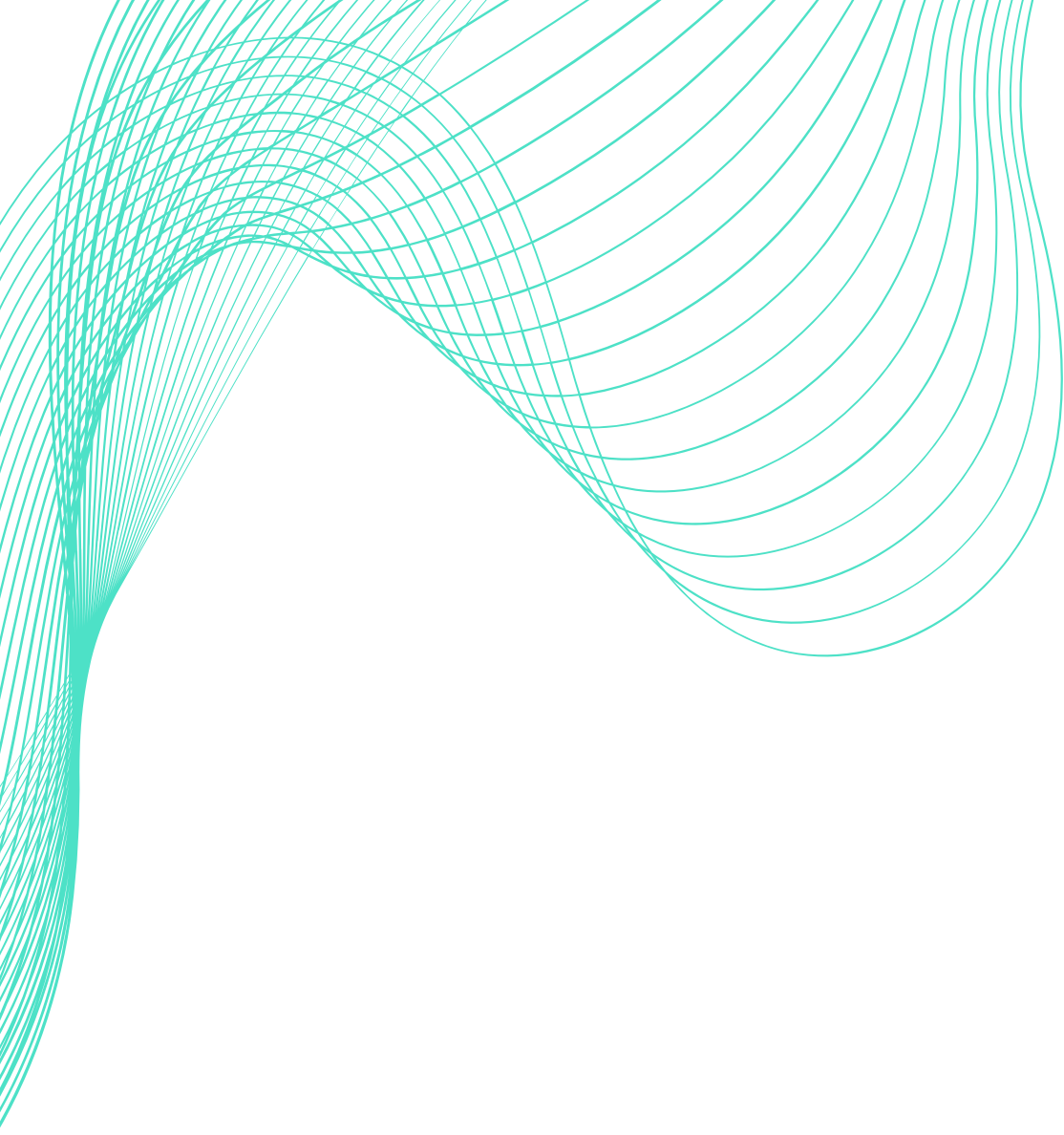
Diversity-Aware Aggregation in Random Forests

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Introduction

- Many trees within a random forest become highly correlated, especially when trained on similar samples or same features are near root
- Leads to redundant votes and limits gains from ensemble learning
- Our work considers tree diversity when aggregating predictions
 - Uses **structural similarity** and **prediction agreement** to measure diversity
- Each tree is assigned a diversity score reflecting its redundancy
 - Weighted voting system



Related Works

Liu et al.

- Built complete random trees to increase tree diversity
- found bootstrap sampling can limit diversity.

Abellan et al.

- Used imprecise probabilities as split criterion
- Increases structural diversity in tree construction

- Prior approaches include: subsampling, anomaly pruning, imprecise probabilities, and genetic algorithms
- Our work explores a novel approach: enhancing tree diversity after training through weighted aggregation at the ensemble stage

Dataset + Preprocessing

Student Performance dataset from Kaggle

Ranked attributes:

0.25235	13	last_exam_score
0.14556	14	assignment_scores_avg
0.12141	16	concept_understanding_score
0.03134	10	ai_generated_content_percentage
0.01656	21	tutoring_hours
0.01535	19	sleep_hours
0.01077	9	ai_dependency_score
0.01062	7	ai_tools_used
0.00956	17	study_consistency_index
0.00909	12	ai_ethics_score
0.00794	3	grade_level
0.00743	15	attendance_percentage
0.00727	11	ai_prompts_per_week
0.00636	4	study_hours_per_day
0.00623	8	ai_usage_purpose
0.00585	18	improvement_rate
0.0052	5	uses_ai
0.00433	1	age
0.00357	20	social_media_hours
0.00347	22	class_participation_score
0.00228	2	gender
0.00106	6	ai_usage_time_minutes

Dataset:

- 8,000 student instances with 25 attributes
- Class variable: *performance_category* ('low', 'medium', 'high')

Preprocessing:

- *student_id* was removed – no predictive information
- *final_score* and *passed* removed – directly indicate performance
- No missing values
- Correlation Attribute Eval with the Ranker search (cutoff =0.008)
- 70% (5,600 instances) training, 15% (1,200 instances) validation, and 15% (1,200 instances) testing

Random Forest Models

Standard Majority Voting

- Baseline model
- Ensemble of 100 decision trees, each trained on the same data
- Scikit-learn
- Trees use maximum depth 10, entropy criterion, class-balanced weights
- Final prediction obtained by uniform voting, each tree has an equal vote

Structural Similarity Aggregation

- Weights each tree based on structural diversity
- Diversity score
 - Record how many times a feature is used as a split in the tree
 - Produce vector summarizing tree structure
 - Compare vectors using cosine similarity
- Weighted sum of tree probabilities used for final prediction

$$\hat{f}_i = \frac{f_i}{\sum_{k=1}^d c_{ik}}$$

Prediction Agreement Aggregation

- Weights each tree based on prediction diversity
- Diversity score
 - Computed from pairwise prediction agreement between trees
 - Fraction of validation samples on which they predict the same class
- Weighted sum of tree probabilities used for final prediction

$$A_{ij} = \frac{1}{M} \sum_{m=1}^M \mathbf{1}[y_i(x_m) = y_j(x_m)]$$

Results/Discussion

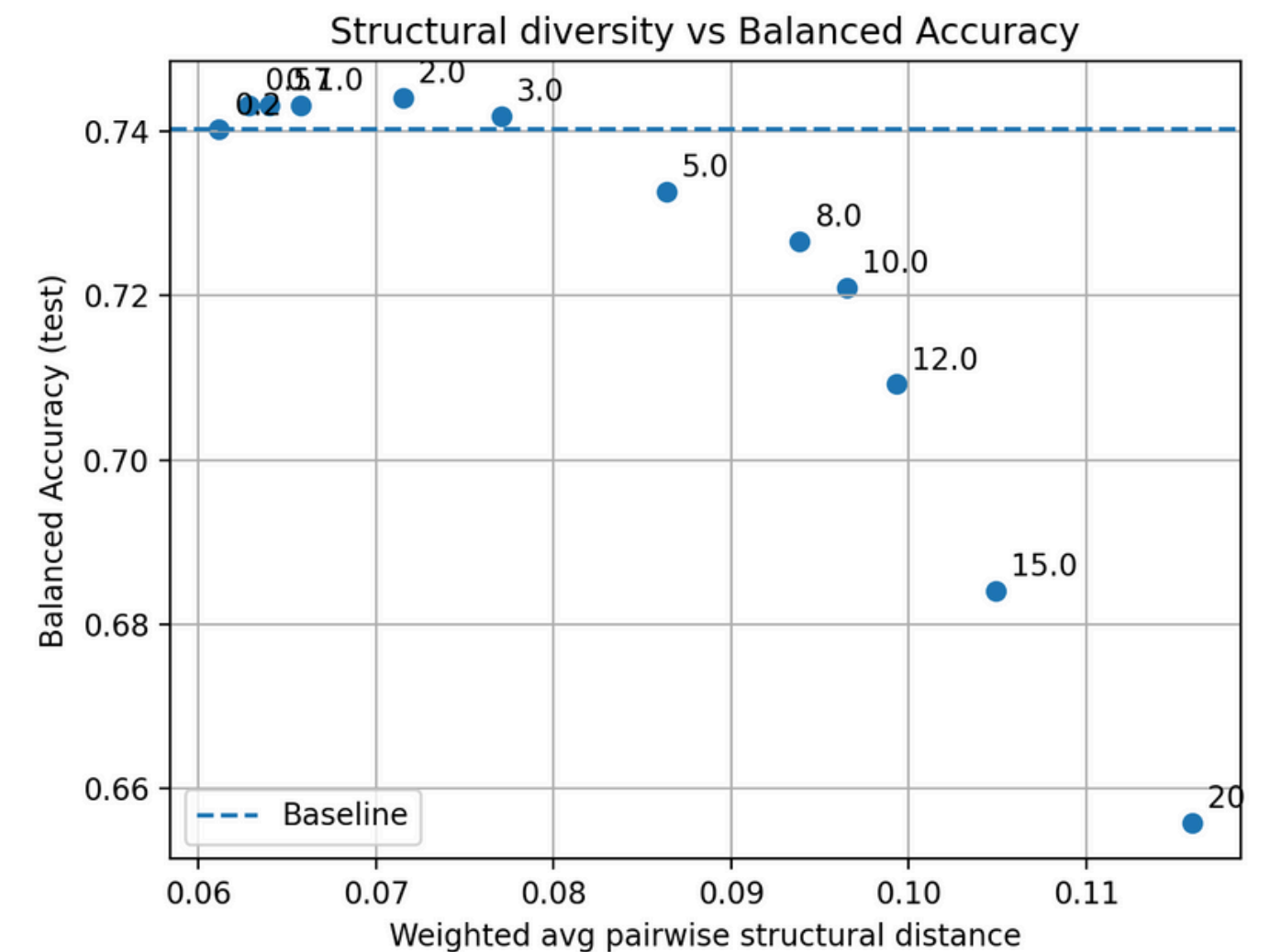
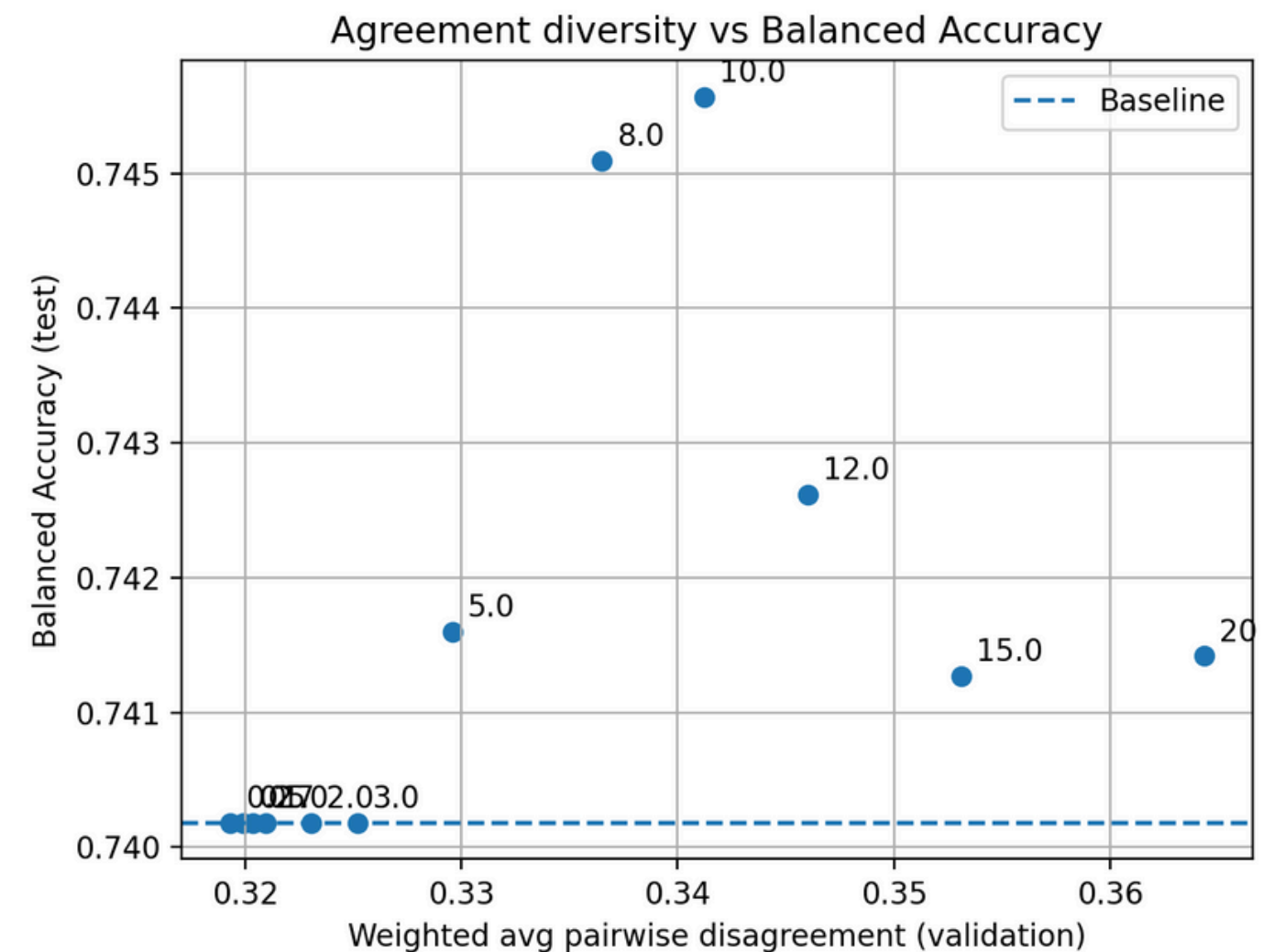
- Structural-weighted RF had highest accuracy (0.7667)
- Improvements are most notable in balanced accuracy, F1-scores, and Cohen's Kappa
 - Indicates more evenly distributed performance across classes, not just majority

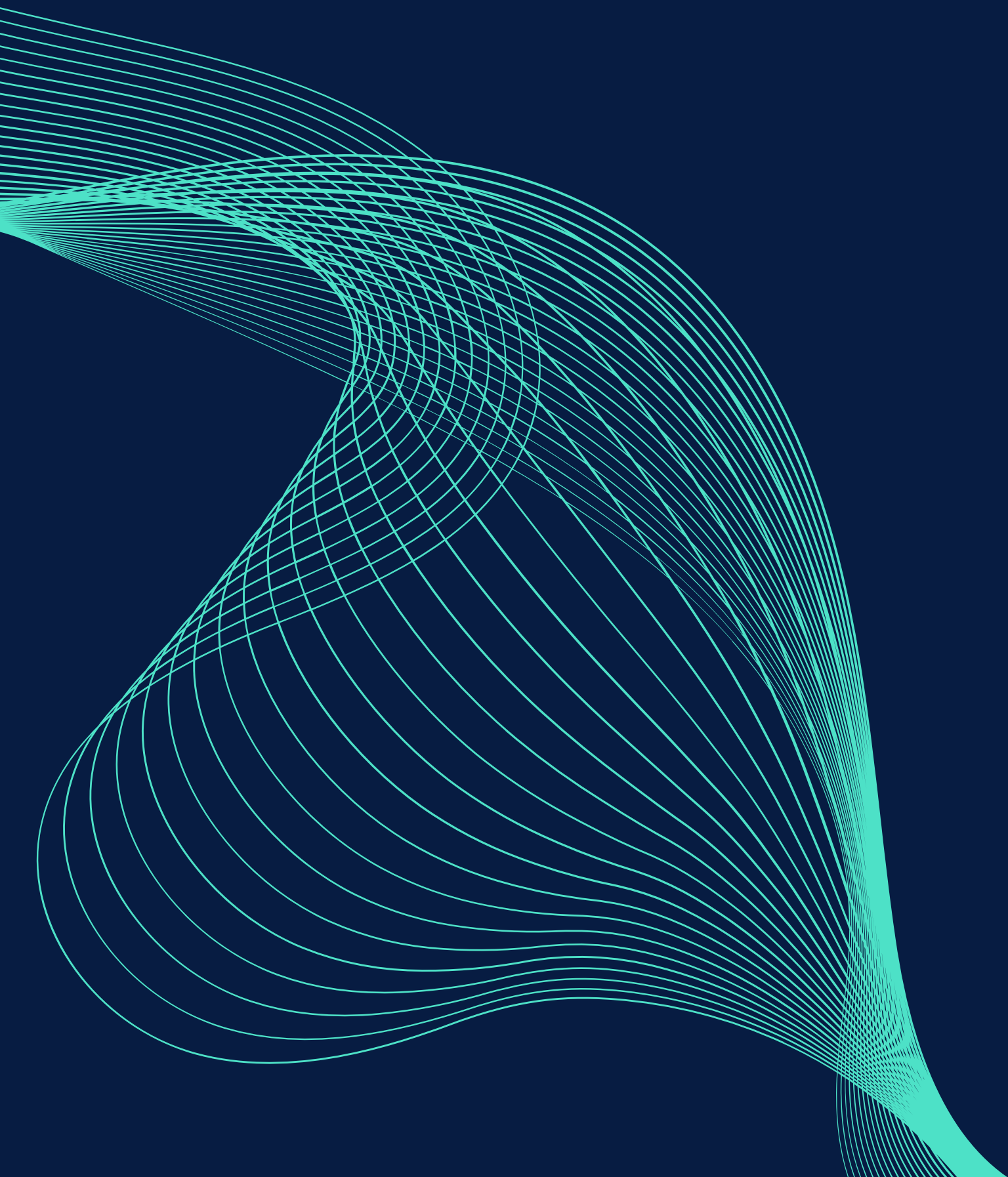
Metric	Baseline RF	Agreement-Weighted RF	Structural-Weighted RF
Accuracy	0.7658	0.7658	0.7667
Balanced Accuracy	0.7355	0.7359	0.7364
Macro F1	0.7167	0.7164	0.7170
Weighted F1	0.7677	0.7678	0.7687
Macro Precision	0.7026	0.7018	0.7025
Macro Recall	0.7355	0.7359	0.7364
Cohen Kappa	0.5829	0.5834	0.5847
Log Loss	0.5192	0.5196	0.5204

Performance Metrics

Results/Discussion

- Analyze how changing α (diversity weight strength) affects performance
- Prediction agreement forest: peak balanced accuracy with moderate α (~8-10)
 - More stable
- Structural diversity forest: very large α degrades performance
- Over sharpening weights collapses ensemble and makes generalization worse





Conclusion and Future Work

- Diversity-aware weighting produced small but consistent improvements over uniform voting
 - Improved class-balanced predictions
- Agreement-based diversity was more reliable than structural diversity
- Future work would focus on the combination of multiple diversity metrics into a single weighted strategy